



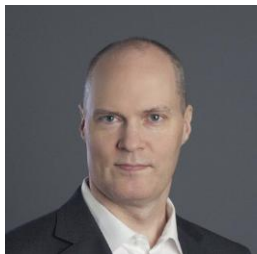
Building Decentralized AI: Best Practices by ArcBlock for
Innovators & Changemakers

Decentralized AI Day (March, 2025)

What's Ahead

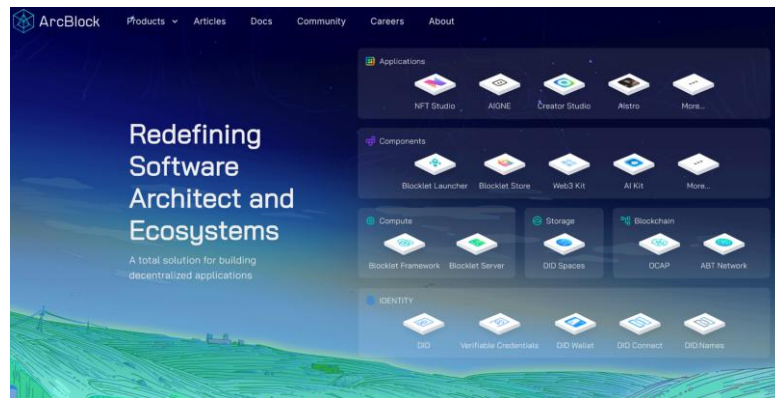
- Ground rules
- What is Decentralized AI
- Benefits and Challenges
- The Do's and Don'ts
- Examples
- What's next (Model Context Protocol)

You Decentralized AI Host



Matt McKinney, ArcBlock

- DIF DID Method Co-chair, Builder, Mentor
- Shaping Global Decentralized AI
- www.aigne.io,
<https://github.com/AIGNE-io/aigne-framework>



“Why did the decentralized AI break up with the central server?”

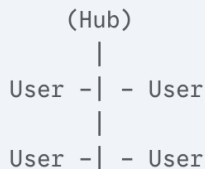
“What did the centralized AI say at the party?”

What is Decentralized AI?

- **DeAI:** AI where users control data and decisions—blockchain for trust, DID for ownership.
- **Key Tech:** Blockchain, Decentralized Identity (DID)
- **Unlike Traditional AI:** No central gatekeepers

Centralized AI

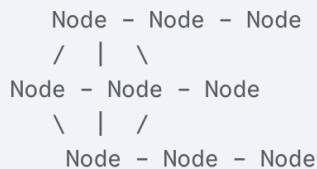
[Corporate Farm]



(Single Point)

Decentralized AI

[Community Garden]



(Interconnected Web)

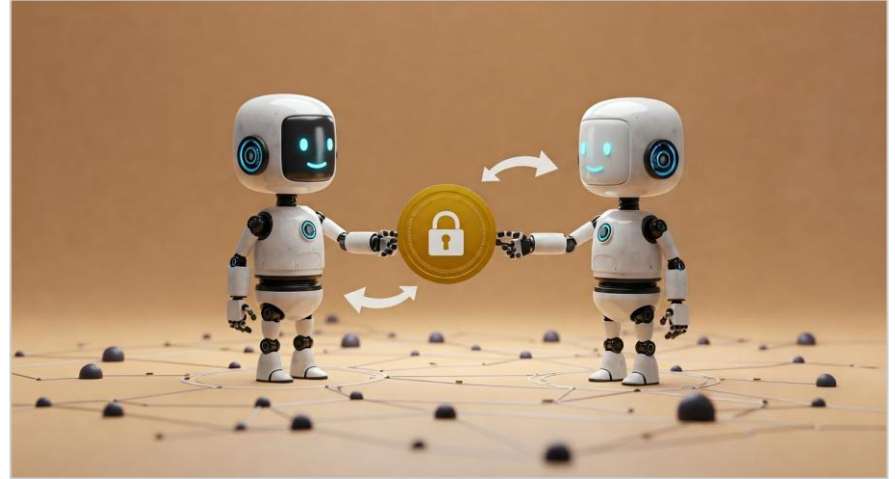
"In DeAI, you plant, harvest like a community garden."



ArcBlock

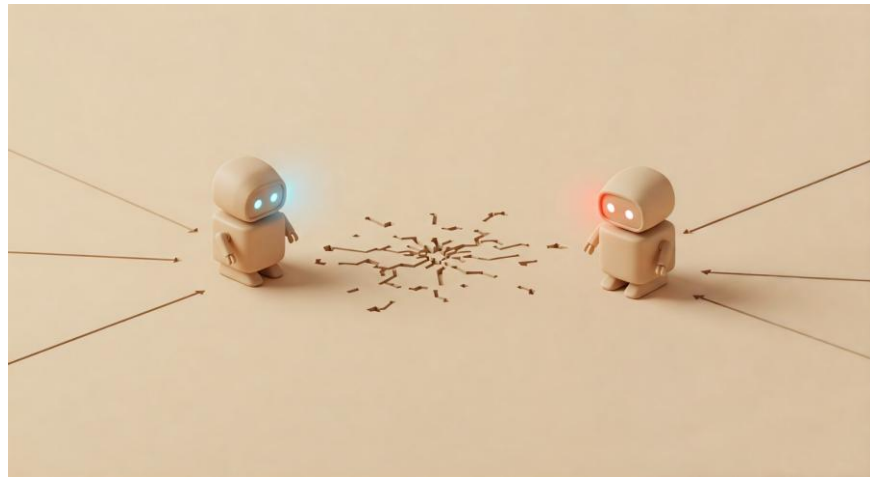
Benefits of Decentralized AI

- **Security:** Limit single points of failure
- **Privacy:** You own your data with DID
- **Trust:** Transparent via immutable records
- **Access:** Levels the playing field for all
- **Resilience:** Keeps running, adapts to you
- **Tokenization:** enables AI-to-AI interactions.



Challenges in DeAI

- **Scalability:** Large networks need power
- **Interoperability:** Connecting diverse systems
- **Data Quality:** Ensuring diverse, clean inputs
- **Regulation:** Unclear legal rules
- **Adoption:** Teaching users the value



Constraint Triangle Comparison



Constraint	Centralized AI	Decentralized AI
Speed	Fast, like a drive-thru	Nearly as fast, optimized nodes. Scale with MCP
Cost	Free, but you pay with your data	Low, most costs covered by company (e.g. ArcBlock)
Response Quality	Decent, but generic	Better, diverse data cuts bias

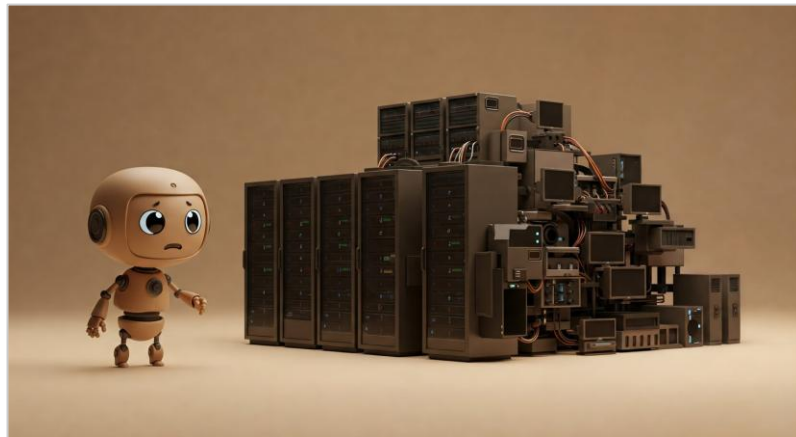
Do's of DeAI

- **Empower Users:** DID for data ownership
- **Be Transparent:** Blockchain logs decisions
- **Stay Safe:** Secure data backups
- **Collaborate:** Reward community input
- **Integrate:** Use open standards
- **Keep Fair:** Diverse data cuts bias



Don'ts of DeAI

- **Skip Governance:** Chaos creeps in
- **Centralize:** Avoid node dominance
- **Sacrifice UX:** Keep it simple
- **Lock In:** Shun proprietary traps
- **Skimp Security:** Build robust defenses
- **Chase Hype:** Solve real problems



Applications and Examples

- **Healthcare:** Private patient data analysis
- **Finance:** Clear credit scoring
- **Supply Chain:** Tamper-proof tracking
- **ArcBlock Examples:**
 - Translation Agent: Fast, no-code translations
 - Learning Plan: Custom education paths
 - Meeting Summarizer: Auto-tasks from talks
 - Creative Studio: co-create art, own it
 - Knowledge Market: share skill, earn rewards



Translation Agent

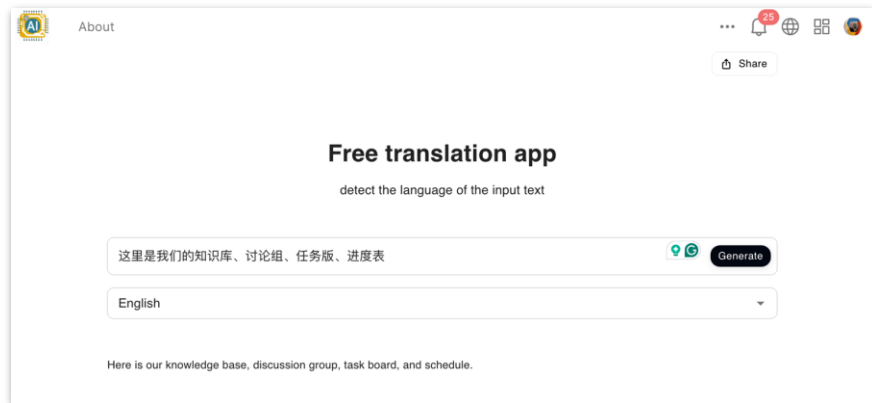
A simple, but efficient, translation app that utilizes a 2 agent configuration to translate and delivery high quality translations.

Sample prompts + guide:

<https://www.arcblock.io/content/blog/build-ai-translation-app-with-no-code>

Preview:

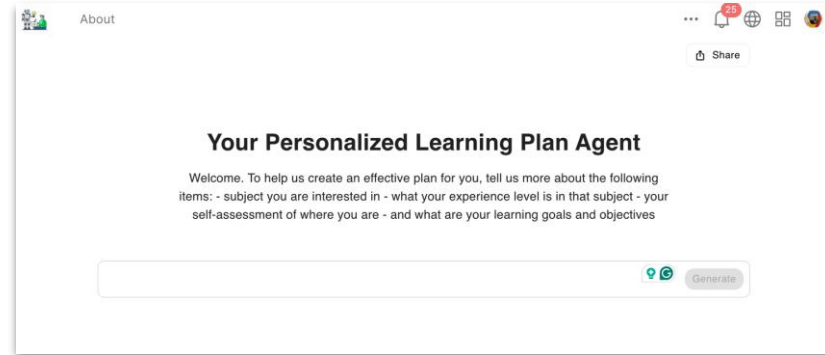
<https://trellex.did.life/apps/540266121962455040>



Learning Plan Agent

An AI agent to create personalized learning experience, adapting to individual needs and preferences.

- Highlights the importance of user interaction and structured output.
- Utilizes a multi-agent configuration to receive the input, create the plan and share it with the user.



Sample prompts + guide:

<https://www.arcblock.io/blog/en/learning-plan-ai-agent>

Preview agent:

<https://trellex.did.life/apps/547209144357093376>

Meeting Summarizer and Action Item Agent

An multi-agent configuration to review, summarize and assign tasks based on meeting transcriptions.

1. Enforce rules and requirements on meeting summaries and automate tasks such as to do's or next steps.
2. Review the guide: <https://www.arcblock.io/blog/en/no-code-meeting-summarizer-ai-agent>

Preview agent:

<https://trellex.did.life/apps/547198239418351616>

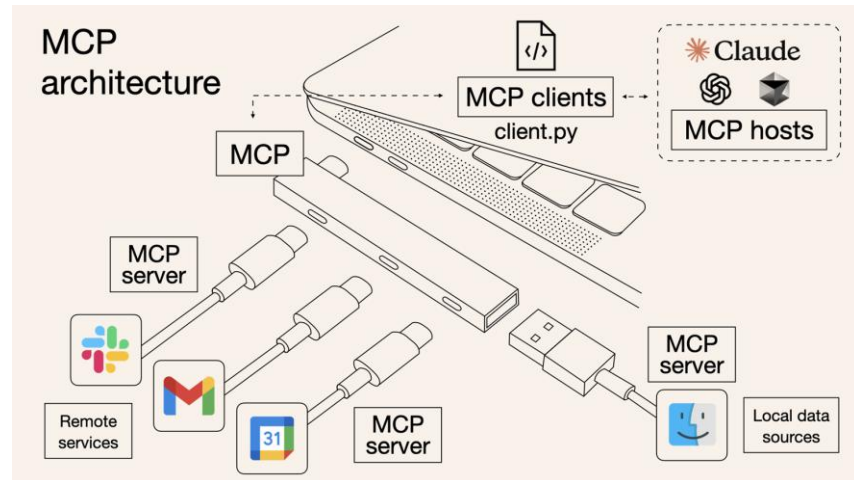
What's Next: Model Context Protocol

A new protocol that will enable DeAI (ie, AIGNE) to connect, use and share with other systems.

- Connects DeAI (e.g. AIGNE) to systems
- Boosts AI with trusted data access

Reference:

<https://www.arcblock.io/content/blog/en/introduction-to-model-context-protocol>



Conclusion and Q&A

- DeAI: Secure, transparent, user-first
- Build smart: Control, scale, collaborate
- Q&A

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